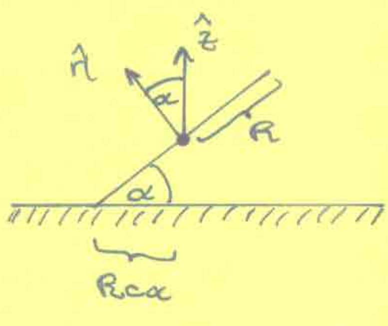
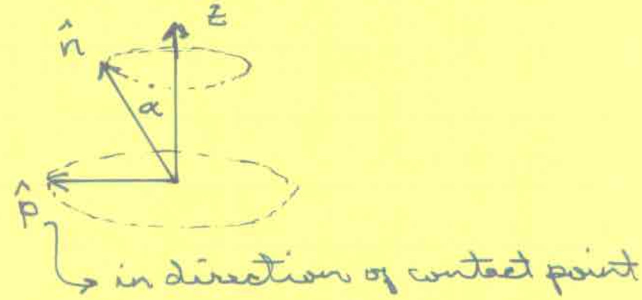
Euler's Disk

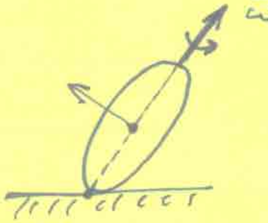


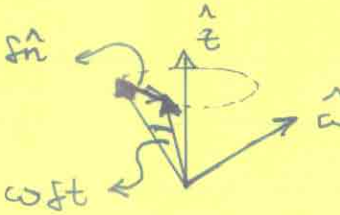
- CM stationary
- contact point traces a circle with radius $Rc\alpha$ at frequency Ω



$$\hat{n} = c\alpha\,\hat{z} + s\alpha\,\hat{p}$$

Since CM is stationary, the infinitesimal motion in time δt is a pure rotation about the axis line passing through contact point and the disk center. Let this angular velocity be $\bar{\omega}$.





$$\hat{\omega} = (\hat{z} - c\alpha\,\hat{n})/s\alpha$$

$$\delta\hat{n} = s\alpha\,\delta\hat{p} \quad ; \quad \delta\hat{p} = \Omega\delta t\,(\hat{z}\times\hat{n})/s\alpha$$

$$= \omega\delta t\,(\hat{z}\times\hat{n})/s\alpha$$

$$\Rightarrow \quad \omega = \Omega s\alpha \quad ; \quad \bar{\omega} = \Omega(\hat{z} - c\alpha\,\hat{n})$$

The above relations result from the geometrical constraints on the motion. We now need to incorporate the dynamics.

The gravitational force produces a torque about the contact point:

$$\bar{N} = R\hat{\omega} \times (-Mg)\hat{z} = -MgR\, c\alpha\, \underbrace{[(\hat{\omega}\times\hat{z})/c\alpha]}_{\equiv(\hat{z}\times\hat{n})/s\alpha} .$$

The instantaneous angular momentum is

$$\bar{h} = I\bar{\omega} \;; \qquad I = \tfrac{1}{4}MR^2$$ (rotation around diameter)

$$\Rightarrow \frac{d\bar{h}}{dt} = I\frac{d\bar{\omega}}{dt} = I(-\Omega c\alpha)\frac{d\hat{\omega}}{dt}$$
$$= -\omega\Omega I c\alpha(\hat{z}\times\hat{n})/s\alpha$$

$$\frac{d\bar{h}}{dt} = \bar{N} \;\Rightarrow\; \omega\Omega I \cancel{c\alpha} = MgR\cancel{c\alpha}$$

$$\Rightarrow \boxed{\Omega^2 s\alpha = 4g/R} .$$

Total energy: $E = \frac{1}{2}\omega^2 I + MgRs\alpha$

$$= \frac{1}{2}I\overbrace{(\Omega^2 s\alpha)}^{4g/R}s\alpha + MgRs\alpha = \frac{1}{2}MgRs\alpha + MgRs\alpha$$
$$= \boxed{\tfrac{3}{2}MgRs\alpha} .$$